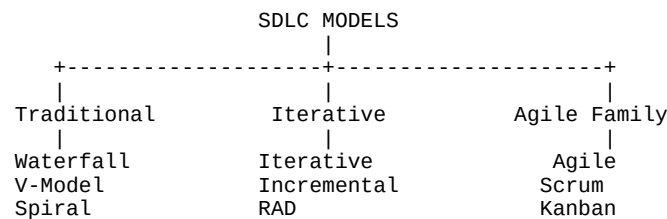


Chapter 2 – Software Development Models: Overview with Diagrams

Overview

A Software Development Life Cycle (SDLC) model defines the sequence of activities used to build software. Different models suit different business needs depending on project size, risk, customer involvement, compliance, and changing requirements.

Classification of SDLC Models



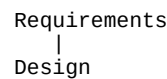
Generic SDLC Lifecycle



Major SDLC Models

Model	Approach	Best For
Waterfall	Sequential	Stable requirements
V-Model	Verification & Validation	Healthcare, Banking
Incremental	Multiple releases	Enterprise systems
Iterative	Continuous refinement	Changing requirements
Spiral	Risk driven	Large complex projects
RAD	Rapid prototyping	Business applications
Agile	Iterative & Incremental	Dynamic products

Waterfall Model





V-Model Model

```
Requirements --> Acceptance Test
System Design --> System Test
Architecture --> Integration Test
Module Design --> Unit Test
Coding
```

Incremental Model

Release 1 --> Release 2 --> Release 3 --> Final Product

Iterative Model

Iteration 1 --> Feedback --> Iteration 2 --> Feedback --> Final

Spiral Model

Planning -> Risk -> Engineering -> Evaluation -> Repeat

RAD Model

Business -> Data -> Process -> App Generation -> Testing

Agile Model

Backlog -> Sprint -> Build -> Test -> Review -> Retro -> Next Sprint

Model Selection Guide

